		↓ ?			Franks. ↓ same index both		minimum ref. ↓ one index only
A_1	16	16	16	16	16	16	16
A_2	16	16	10	6	16	14	6 (10)
I_1	16	7	10	12	14	16	(24) (12) etc.
I_2	16	7	10	12	—	—	—
	64	46					
	X						

assume geom. addr. of Indices

questions: (1) range of A_2. does it need to be whole mem?
(2) separate I_2, needed or not.
(3) 2 oprn arithmetic? (how imp.)

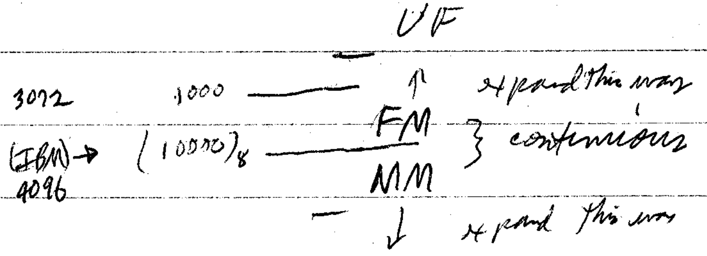
---

1. no. system
2. arith.
3. addr.

2

200,000	2^{16}	65536
100,000	2^{15}	32768
40,000	2^{14}	16384
20,000	2^{13}	8192
10,000	2^{12}	4096
4000	2^{11}	2048
2000	2^{10}	1024
1000	2^{9}	512
400	2^{8}	256
0200	2^{7}	128
100	2^{6}	
40	2^{5}	
20	2^{4}	
10	2^{3}	
4	2^{2}	
2	2^{1}	
0	2^{0}	

stretch addressing



MM in blocks of $(20,000)_8$

A

I

π { P (pre load)
Pr (Pre-store)
Po (post store)
Px (pre-load & post store) same place

*

A I π

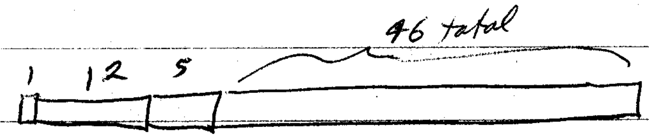
A I I

or going one step further

A (,) π

can add indices, not just one,
each bit refers to 1 indx reg.
giving several, means add all.

→ { go thru example of store with relative addressing too
+ see how it looks —



would like

$A_1 = 16$ bits total mem.

$I_1 =$ [illegible] 16 16 fast regs 1 bit each

$A_2 = 16$

$I_2 =$ [illegible] 16 16 fast regs 1 bit each.

64

$A^0_1 = 16$ [illegible]

$I_1 = 7$

$A_2 = 16$

$I_2 = 7$

46

$A_1 = 16$

$I_1 = 12$

$A_2 = 16$

$I_2 = 12$

56

one scheme.

$A_1 = 16$ MM

$I_1 = 10$

$A_2 = 10$

$I_2 = 10$ (1024)

46

$A_1 = 16$

$A_2 = 9$ 6 UF

$I_1 = 10 \rightarrow 12$?

$I_2 = 10 \rightarrow 8$?

Main Mem. A_1 16

A_2 10 [illegible] 6 (64 / 256 / 1024 [illegible]) all UF & most of FM [illegible]

geom. I_1 [illegible] 10 12 10 of UF

geom. I_2 [illegible] 10 12 same 10 of UF

46

go into question of whether I_2 is necessary ??

29
30
20
79

```
ca   δ
SAb  |y|
TM   c ( x if)
CA   y^n-1
TZ

CA   X
S
St   X-X
CA
9
ST   y-y
- LQ
FA   y
Di   ey
- SQ
CA
     add ub
     T P
     ca
  -> add
     T - (c) repeat
     CA  AXo
     J
```

24
18
11

28
31
30
20
109

28
7

1

30
8

28
-13
15

28
-18
13

10

38
21
7

32
-18
14

20
13
7

29
18
11

28
13
15

30
20
10

16
11

28
28
32
20
108

32
20
12

28
15
13

19
15
4
6

18
15
3

20
15
5

17